

$$\mathbf{ACI} = (\text{magnitude_s1} + \text{frequency_s1} + \text{duration_s1}) + (\text{magnitude_s2} + \text{frequency_s2} + \text{duration_s2})$$

$$\mathbf{SCI} = \mathbf{ACI} + ((\text{magnitude_s1} \times \text{magnitude_s2}) + (\text{frequency_s1} \times \text{frequency_s2}) + (\text{duration_s1} \times \text{duration_s2}))$$

$$\mathbf{ANCI} = \mathbf{ACI} - ((\text{magnitude_s1} \times \text{magnitude_s2}) + (\text{frequency_s1} \times \text{frequency_s2}) + (\text{duration_s1} \times \text{duration_s2}))$$

...and so on between stressors 1,2,3,...etc.

Then, once all matrix values are calculated, sum either the rows or columns for each pollutant type for the final stressor cumulative impact for points ranking based system.

Total stressor interaction Impact =

Additive Stressor Interaction Cumulative Impact (**ACI**) +
Synergistic Stressor Interaction Cumulative Impact (**SCI**) +
(- Antagonistic Stressor Interaction Cumulative Impact (**ANCI**))

$$\text{Toxaphene on itself} = 131.68 + 2387.306 + -2123.946 = 395.04$$

$$\mathbf{ACI} = (44.84 + 7 + 14) + (44.84 + 7 + 14) = 131.68$$

$$\mathbf{SCI} = 131.68 + (44.84 \times 44.84 + 7 \times 7 + 14 \times 14) = 2387.306$$

$$\mathbf{ANCI} = 131.68 - (44.84 \times 44.84 + 7 \times 7 + 14 \times 14) = -2123.946$$

$$\text{PCBs on itself} = 211.68 + 7670.506 + -7247.146 = 635.04$$

$$\mathbf{ACI} = (84.84 + 6 + 15) + (84.84 + 6 + 15) = 211.68$$

$$\mathbf{SCI} = 211.68 + (84.84 \times 84.84 + 6 \times 6 + 15 \times 15) = 7670.506$$

$$\mathbf{ANCI} = 211.68 - (84.84 \times 84.84 + 6 \times 6 + 15 \times 15) = -7247.146$$

$$\text{Mirex on itself} = 123.62 + 2280.556 + -2033.316 = 370.86$$

$$\mathbf{ACI} = (44.81 + 7 + 10) + (44.81 + 7 + 10) = 123.62$$

$$\mathbf{SCI} = 123.62 + (44.81 \times 44.81 + 7 \times 7 + 10 \times 10) = 2280.556$$

$$\mathbf{ANCI} = 123.62 - (44.81 \times 44.81 + 7 \times 7 + 10 \times 10) = -2033.316$$

$$\text{Mercury on itself (up to about 60 days half life)} = 285.628 + 17263.777 + -16692.521 = 856.884$$

$$\mathbf{ACI} = (129.65 + 13 + 0.164) + (129.65 + 13 + 0.164) = 285.628$$

$$\mathbf{SCI} = 285.628 + (129.65 \times 129.65 + 13 \times 13 + 0.164 \times 0.164) = 17263.777$$

$$\mathbf{ANCI} = 285.628 - (129.65 \times 129.65 + 13 \times 13 + 0.164 \times 0.164) = -16692.521$$

$$\text{Heptachlor on itself} = 151.2 + 4653.132 + -4350.732 = 453.6$$

$$\mathbf{ACI} = (66.49 + 9 + 0.11) + (66.49 + 9 + 0.11) = 151.2$$

$$\mathbf{SCI} = 151.2 + (66.49 \times 66.49 + 9 \times 9 + 0.11 \times 0.11) = 4653.132$$

$$\mathbf{ANCI} = 151.2 - (66.49 \times 66.49 + 9 \times 9 + 0.11 \times 0.11) = -4350.732$$

$$\text{Fecal Coliform on itself} = 26.946 + 162.507 + -108.615 = 80.838$$

$$\mathbf{ACI} = (11.47 + 2 + 0.003) + (11.47 + 2 + 0.003) = 26.946$$

$$\mathbf{SCI} = 26.946 + (11.47 \times 11.47 + 2 \times 2 + 0.003 \times 0.003) = 162.507$$

$$\mathbf{ANCI} = 26.946 - (11.47 \times 11.47 + 2 \times 2 + 0.003 \times 0.003) = -108.615$$

$$\text{Endrin on itself} = 131.62 + 2384.556 + -2121.316 = 394.86$$

$$\mathbf{ACI} = (44.81 + 7 + 14) + (44.81 + 7 + 14) = 131.62$$

$SCI = 131.62 + (44.81 \times 44.81 + 7 \times 7 + 14 \times 14) = 2384.556$
 $ANCI = 131.62 - (44.81 \times 44.81 + 7 \times 7 + 14 \times 14) = -2121.316$
 Dieldrin on itself = $160.98 + 4687.9 + -4365.94 = 482.94$
 $ACI = (66.49 + 9 + 5) + (66.49 + 9 + 5) = 160.98$
 $SCI = 160.98 + (66.49 \times 66.49 + 9 \times 9 + 5 \times 5) = 4687.9$
 $ANCI = 160.98 - (66.49 \times 66.49 + 9 \times 9 + 5 \times 5) = -4365.94$
 Aldrin on itself = $1988.78 + 957428.992 - 953451.432 = 5966.34$
 $ACI = (977.39 + 7 + 10) + (977.39 + 7 + 10) = 1988.78$
 $SCI = 1988.78 + (977.39 \times 977.39 + 7 \times 7 + 10 \times 10) = 957428.992$
 $ANCI = 1988.78 - (977.39 \times 977.39 + 7 \times 7 + 10 \times 10) = -953451.432$

Toxaphene on PCBs = $171.68 + 4227.906 + -3884.546 =$
 $ACI = (44.84 + 7 + 14) + (84.84 + 6 + 15) = 171.68$
 $SCI = 171.68 + (44.84 \times 84.84 + 7 \times 6 + 14 \times 15) = 4227.906$
 $ANCI = 171.68 - (44.84 \times 84.84 + 7 \times 6 + 14 \times 15) = -3884.546$
 Toxaphene on Mirex = $127.65 + 2325.93 + -2070.63 = 382.95$
 $ACI = (44.84 + 7 + 14) + (44.81 + 7 + 10) = 127.65$
 $SCI = 127.65 + (44.84 \times 44.81 + 7 \times 7 + 14 \times 10) = 2325.93$
 $ANCI = 127.65 - (44.84 \times 44.81 + 7 \times 7 + 14 \times 10) = -2070.63$
 Toxaphene on Mercury = $208.654 + 6115.456 + -5698.148 = 625.962$
 $ACI = (44.84 + 7 + 14) + (129.65 + 13 + 0.164) = 208.654$
 $SCI = 208.654 + (44.84 \times 129.65 + 7 \times 13 + 14 \times 0.164) = 6115.456$
 $ANCI = 208.654 - (44.84 \times 129.65 + 7 \times 13 + 14 \times 0.164) = -5698.148$
 Toxaphene on Heptachlor = $141.44 + 3187.3916 + -2904.512 = 424.320$
 $ACI = (44.84 + 7 + 14) + (66.49 + 9 + 0.11) = 141.44$
 $SCI = 141.44 + (44.84 \times 66.49 + 7 \times 9 + 14 \times 0.11) = 3187.3916$
 $ANCI = 141.44 - (44.84 \times 66.49 + 7 \times 9 + 14 \times 0.11) = -2904.512$
 Toxaphene on Fecal Coliform = $79.313 + 607.6698 + -449.0438 = 237.939$
 $ACI = (44.84 + 7 + 14) + (11.47 + 2 + 0.003) = 79.313$
 $SCI = 79.313 + (44.84 \times 11.47 + 7 \times 2 + 14 \times 0.003) = 607.6698$
 $ANCI = 79.313 - (44.84 \times 11.47 + 7 \times 2 + 14 \times 0.003) = -449.0438$
 Toxaphene on Endrin = $131.65 + 2385.93 + -2122.63 = 394.95$
 $ACI = (44.84 + 7 + 14) + (44.81 + 7 + 14) = 131.65$
 $SCI = 131.65 + (44.84 \times 44.81 + 7 \times 7 + 14 \times 14) = 2385.93$
 $ANCI = 131.65 - (44.84 \times 44.81 + 7 \times 7 + 14 \times 14) = -2122.63$
 Toxaphene on Dieldrin = $146.33 + 3260.742 + -2968.082 = 438.99$
 $ACI = (44.84 + 7 + 14) + (66.49 + 9 + 5) = 146.33$
 $SCI = 146.33 + (44.84 \times 66.49 + 7 \times 9 + 14 \times 5) = 3260.742$
 $ANCI = 146.33 - (44.84 \times 66.49 + 7 \times 9 + 14 \times 5) = -2968.082$
 Toxaphene on Aldrin = $1060.23 + 45075.398 + -42954.938 = 3180.69$
 $ACI = (44.84 + 7 + 14) + (977.39 + 7 + 10) = 1060.23$
 $SCI = 1060.23 + (44.84 \times 977.39 + 7 \times 7 + 14 \times 10) = 45075.398$
 $ANCI = 1060.23 - (44.84 \times 977.39 + 7 \times 7 + 14 \times 10) = -42954.938$

PCBs on Mirex = $167.65 + 4161.33 + -3826.03 = 502.95$
 $ACI = (84.84 + 6 + 15) + (44.81 + 7 + 10) = 167.65$
 $SCI = 167.65 + (84.84 \times 44.81 + 6 \times 7 + 15 \times 10) = 4161.33$
 $ANCI = 167.65 - (84.84 \times 44.81 + 6 \times 7 + 15 \times 10) = -3826.03$
 PCBs on Mercury = $248.654 + 11328.62 + -10831.312 = 745.962$
 $ACI = (84.84 + 6 + 15) + (129.65 + 13 + 0.164) = 248.654$
 $SCI = 248.654 + (84.84 \times 129.65 + 6 \times 13 + 15 \times 0.164) = 11328.62$

$ANCI = 248.654 - (84.84 \times 129.65 + 6 \times 13 + 15 \times 0.164) = -10831.312$
 PCBs on Heptachlor = $181.44 + 5878.102 + -5515.222 = 544.32$
 $ACI = (84.84 + 6 + 15) + (66.49 + 9 + 0.11) = 181.44$
 $SCI = 181.44 + (84.84 \times 66.49 + 6 \times 9 + 15 \times 0.11) = 5878.102$
 $ANCI = 181.44 - (84.84 \times 66.49 + 6 \times 9 + 15 \times 0.11) = -5515.222$
 PCBs on Fecal Coliform = $119.313 + 1104.4728 + -865.8468 = 357.939$
 $ACI = (84.84 + 6 + 15) + (11.47 + 2 + 0.003) = 119.313$
 $SCI = 119.313 + (84.84 \times 11.47 + 6 \times 2 + 15 \times 0.003) = 1104.4728$
 $ANCI = 119.313 - (84.84 \times 11.47 + 6 \times 2 + 15 \times 0.003) = -865.8468$
 PCBs on Endrin = $171.65 + 4225.3304 + -3882.0304 = 514.95$
 $ACI = (84.84 + 6 + 15) + (44.81 + 7 + 14) = 171.65$
 $SCI = 171.65 + (84.84 \times 44.81 + 6 \times 7 + 15 \times 14) = 4225.3304$
 $ANCI = 171.65 - (84.84 \times 44.81 + 6 \times 7 + 15 \times 14) = -3882.0304$
 PCBS on Dieldrin = $186.33 + 5956.3416 + -5583.6816 = 558.99$
 $ACI = (84.84 + 6 + 15) + (66.49 + 9 + 5) = 186.33$
 $SCI = 186.33 + (84.84 \times 66.49 + 6 \times 9 + 15 \times 5) = 5956.3416$
 $ANCI = 186.33 - (84.84 \times 66.49 + 6 \times 9 + 15 \times 5) = -5583.6816$
 PCBS on Aldrin = $1100.23 + 84213.998 + -82013.538 = 3300.69$
 $ACI = (84.84 + 6 + 15) + (977.39 + 7 + 10) = 1100.23$
 $SCI = 1100.23 + (84.84 \times 977.39 + 6 \times 7 + 15 \times 10) = 84213.998$
 $ANCI = 1100.23 - (84.84 \times 977.39 + 6 \times 7 + 15 \times 10) = -82013.538$

Mirex on Mercury = $204.624 + 6106.8805 + -5697.6325 = 613.872$
 $ACI = (44.81 + 7 + 10) + (129.65 + 13 + 0.164) = 204.624$
 $SCI = 204.624 + (44.81 \times 129.65 + 7 \times 13 + 10 \times 0.164) = 6106.8805$
 $ANCI = 204.624 - (44.81 \times 129.65 + 7 \times 13 + 10 \times 0.164) = -5697.6325$
 Mirex on Heptachlor = $137.41 + 3180.9269 + -2906.107 = 412.2299$
 $ACI = (44.81 + 7 + 10) + (66.49 + 9 + 0.11) = 137.41$
 $SCI = 137.41 + (44.81 \times 66.49 + 7 \times 9 + 10 \times 0.11) = 3180.9269$
 $ANCI = 137.41 - (44.81 \times 66.49 + 7 \times 9 + 10 \times 0.11) = -2906.107$
 Mirex on Fecal Coliform = $75.283 + 603.2837 + -452.7177 = 225.849$
 $ACI = (44.81 + 7 + 10) + (11.47 + 2 + 0.003) = 75.283$
 $SCI = 75.283 + (44.81 \times 11.47 + 7 \times 2 + 10 \times 0.003) = 603.2837$
 $ANCI = 75.283 - (44.81 \times 11.47 + 7 \times 2 + 10 \times 0.003) = -452.7177$
 Mirex on Endrin = $127.62 + 2324.5561 + -2069.316 = 382.8601$
 $ACI = (44.81 + 7 + 10) + (44.81 + 7 + 14) = 127.62$
 $SCI = 127.62 + (44.81 \times 44.81 + 7 \times 7 + 10 \times 14) = 2324.5561$
 $ANCI = 127.62 - (44.81 \times 44.81 + 7 \times 7 + 10 \times 14) = -2069.316$
 Mirex on Dieldrin = $142.3 + 3234.7169 + -2950.117 = 426.8999$
 $ACI = (44.81 + 7 + 10) + (66.49 + 9 + 5) = 142.3$
 $SCI = 142.3 + (44.81 \times 66.49 + 7 \times 9 + 10 \times 5) = 3234.7169$
 $ANCI = 142.3 - (44.81 \times 66.49 + 7 \times 9 + 10 \times 5) = -2950.117$
 Mirex on Aldrin = $1056.2 + 45002.0459 + -42889.646 = 3168.5999$
 $ACI = (44.81 + 7 + 10) + (977.39 + 7 + 10) = 1056.2$
 $SCI = 1056.2 + (44.81 \times 977.39 + 7 \times 7 + 10 \times 10) = 45002.0459$
 $ANCI = 1056.2 - (44.81 \times 977.39 + 7 \times 7 + 10 \times 10) = -42889.646$

Mercury on Heptachlor = $218.414 + 8955.86054 + -8519.03254 = 655.242$

$ACI = (129.65 + 13 + 0.164) + (66.49 + 9 + 0.11) = 218.414$
 $SCI = 218.414 + (129.65 \times 66.49 + 13 \times 9 + 0.164 \times 0.11) = 8955.86054$
 $ANCI = 218.414 - (129.65 \times 66.49 + 13 \times 9 + 0.164 \times 0.11) = -8519.03254$
 Mercury on Fecal Coliform = $156.287 + 1669.372992 + -1356.798992 = 468.861$
 $ACI = (129.65 + 13 + 0.164) + (11.47 + 2 + 0.003) = 156.287$
 $SCI = 156.287 + (129.65 \times 11.47 + 13 \times 2 + 0.164 \times 0.003) = 1669.372992$
 $ANCI = 156.287 - (129.65 \times 11.47 + 13 \times 2 + 0.164 \times 0.003) = -1356.798992$
 Mercury on Endrin = $208.624 + 6111.5365 + -5694.2885 = 625.872$
 $ACI = (129.65 + 13 + 0.164) + (44.81 + 7 + 14) = 208.624$
 $SCI = 208.624 + (129.65 \times 44.81 + 13 \times 7 + 0.164 \times 14) = 6111.5365$
 $ANCI = 208.624 - (129.65 \times 44.81 + 13 \times 7 + 0.164 \times 14) = -5694.2885$
 Mercury on Dieldrin = $223.304 + 8961.5525 + -8514.9445 = 669.912$
 $ACI = (129.65 + 13 + 0.164) + (66.49 + 9 + 5) = 223.304$
 $SCI = 223.304 + (129.65 \times 66.49 + 13 \times 9 + 0.164 \times 5) = 8961.5525$
 $ANCI = 223.304 - (129.65 \times 66.49 + 13 \times 9 + 0.164 \times 5) = -8514.9445$
 Mercury on Aldrin = $1137.204 + 127948.4575 + -125674.0495 = 3411.612$
 $ACI = (129.65 + 13 + 0.164) + (977.39 + 7 + 10) = 1137.204$
 $SCI = 1137.204 + (129.65 \times 977.39 + 13 \times 7 + 0.164 \times 10) = 127948.4575$
 $ANCI = 1137.204 - (129.65 \times 977.39 + 13 \times 7 + 0.164 \times 10) = -125674.0495$

Heptachlor on Fecal Coliform = $89.073 + 869.71363 + -691.56763 = 267.219$
 $ACI = (66.49 + 9 + 0.11) + (11.47 + 2 + 0.003) = 89.073$
 $SCI = 89.073 + (66.49 \times 11.47 + 9 \times 2 + 0.11 \times 0.003) = 869.71363$
 $ANCI = 89.073 - (66.49 \times 11.47 + 9 \times 2 + 0.11 \times 0.003) = -691.56763$

Heptachlor on Endrin = $141.41 + 3185.3669 + -2902.547 = 424.2299$
 $ACI = (66.49 + 9 + 0.11) + (44.81 + 7 + 14) = 141.41$
 $SCI = 141.41 + (66.49 \times 44.81 + 9 \times 7 + 0.11 \times 14) = 3185.3669$
 $ANCI = 141.41 - (66.49 \times 44.81 + 9 \times 7 + 0.11 \times 14) = -2902.547$
 Heptachlor on Dieldrin = $156.09 + 4658.56 + -4346.38 = 468.27$
 $ACI = (66.49 + 9 + 0.11) + (66.49 + 9 + 5) = 156.09$
 $SCI = 156.09 + (66.49 \times 66.49 + 9 \times 9 + 0.11 \times 5) = 4658.56$
 $ANCI = 156.09 - (66.49 \times 66.49 + 9 \times 9 + 0.11 \times 5) = -4346.38$
 Heptachlor on Aldrin = $1069.99 + 66120.7511 + -63980.7711 = 3209.97$
 $ACI = (66.49 + 9 + 0.11) + (977.39 + 7 + 10) = 1069.99$
 $SCI = 1069.99 + (66.49 \times 977.39 + 9 \times 7 + 0.11 \times 10) = 66120.7511$
 $ANCI = 1069.99 - (66.49 \times 977.39 + 9 \times 7 + 0.11 \times 10) = -63980.7711$

Fecal Coliform on Endrin = $79.283 + 607.2957 + -448.7297 = 237.849$
 $ACI = (11.47 + 2 + 0.003) + (44.81 + 7 + 14) = 79.283$
 $SCI = 79.283 + (11.47 \times 44.81 + 2 \times 7 + 0.003 \times 14) = 607.2957$
 $ANCI = 79.283 - (11.47 \times 44.81 + 2 \times 7 + 0.003 \times 14) = -448.7297$
 Fecal Coliform on Dieldrin = $93.963 + 874.6183 + -686.6923 = 281.889$
 $ACI = (11.47 + 2 + 0.003) + (66.49 + 9 + 5) = 93.963$
 $SCI = 93.963 + (11.47 \times 66.49 + 2 \times 9 + 0.003 \times 5) = 874.6183$
 $ANCI = 93.963 + (11.47 \times 66.49 + 2 \times 9 + 0.003 \times 5) = -686.6923$
 Fecal Coliform on Aldrin = $1007.863 + 12232.5563 + -10216.8303 = 3023.589$
 $ACI = (11.47 + 2 + 0.003) + (977.39 + 7 + 10) = 1007.863$
 $SCI = 1007.863 + (11.47 \times 977.39 + 2 \times 7 + 0.003 \times 10) = 12232.5563$
 $ANCI = 1007.863 - (11.47 \times 977.39 + 2 \times 7 + 0.003 \times 10) = -10216.8303$

$$\text{Endrin on Dieldrin} = 146.3 + 3258.717 + -2966.117 = 438.9$$

$$\text{ACI} = (44.81 + 7 + 14) + (66.49 + 9 + 5) = 146.3$$

$$\text{SCI} = 146.3 + (44.81 \times 66.49 + 7 \times 9 + 14 \times 5) = 3258.717$$

$$\text{ANCI} = 146.3 - (44.81 \times 66.49 + 7 \times 9 + 14 \times 5) = -2966.117$$

$$\text{Endrin on Aldrin} = 1060.2 + 45046.046 + -42925.646 = 3180.6$$

$$\text{ACI} = (44.81 + 7 + 14) + (977.39 + 7 + 10) = 1060.2$$

$$\text{SCI} = 1060.2 + (44.81 \times 977.39 + 7 \times 7 + 14 \times 10) = 45046.046$$

$$\text{ANCI} = 1060.2 - (44.81 \times 977.39 + 7 \times 7 + 14 \times 10) = -42925.646$$

$$\text{Dieldrin on Aldrin} = 1074.88 + 66174.541 + -64024.781 = 3,224.64$$

$$\text{ACI} = (66.49 + 9 + 5) + (977.39 + 7 + 10) = 1074.88$$

$$\text{SCI} = 1074.88 + (66.49 \times 977.39 + 9 \times 7 + 5 \times 10) = 66174.541$$

$$\text{ANCI} = 1074.88 - (66.49 \times 977.39 + 9 \times 7 + 5 \times 10) = -64024.781$$